

Compared to Quiz 1, the questions in Quiz 2 are more complex and longer in length. 1

Question 1

This question is about the Continuous-Time System State Space. Given the

$y(t) = x_1(t)$, $x_2(t) = \frac{dx_1(t)}{dt}$, $y(t) = x_1(t)$, $x_2(t) = \frac{dx_1(t)}{dt}$, and the relation equations are refers to

$$5\frac{d^2y(t)}{dt^2} - 2\frac{dy(t)}{dt} = 2u(t)$$

$$5\frac{d^2y(t)}{dt^2} - 2\frac{dy(t)}{dt} = 2u(t)$$

(i) Find the Continuous-time system State equations.

(ii) Find the Transfer function $\frac{Y(s)}{U(s)}$.

Question 2

The knowledge it tested is the transform from Continuous time to Discrete time State Space.

Mainly question stem: given

$$\dot{X} = \begin{bmatrix} 1 & \beta \\ 0 & 2 \end{bmatrix} X + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

$$\dot{X} = [1 \ 0 \ \beta \ 2] X + [0 \ 1] u(t)$$

$$Y = [1 \ 0] X$$

$$Y = [1 \ 0] X$$

(i) With the ZOH, convert the Continuous to Discrete space, with sampling time $T=0.5$ second.

Find the final condition.

(ii) Compute the value of $\frac{X(z)}{U(z)}$ with the parameter β .

(iii) Analyze the controllability of discrete system while β varies.

Question 3

Discrete Transition Equation.

For discrete time space model

$$\dot{X} = \begin{bmatrix} 1 & 0.5 \\ -0.5 & -1 \end{bmatrix} X + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u(t)$$

$$\dot{X} = [1 \ -0.5 \ 0.5 \ -1] X + [1 \ 1] u(t)$$

Given the $\begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} [x_1(0) \ x_2(0)] = \begin{bmatrix} 2 \\ 2 \end{bmatrix} [2 \ 2]$, $u(0) = 1$, $u(1) = -1$.

Find the value of $x(2)$